

Denis-Alexandru Hatu

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EXPERIENCE

Junior Data Engineer @Lobyco | *Azure, Databricks, SQL, PySpark* Nov. 2025 – Present

- Provisioned and deployed infrastructure for a **Databricks** Data Platform using **Terraform** and **Azure DevOps** pipelines, coordinating with other core teams.
- Designed and provided maintenance to ETL pipelines using **PySpark** and **JSON** file manipulation.
- Single-handedly fixed several **deployment-related issues**, such as type-precision bugs, filtering errors and missing data.
- Investigated and resolved over **50 production incidents** reported through **Jira**, digging into **Unity Catalog** and **Delta Sharing** logs to trace root cause and shipped hotfixes straight to production under time pressure.

PROJECTS

Chatbot ACS | *Python, Azure, Scrapy, Redis, Docling, LangChain, Streamlit, Claude API*

MultiAgent Energy Community | *Python, Mesa, SQLite, Pandas, NumPy, scikit-learn, Plotly*

- Processed and combined household consumption, solar production, and appliance-level data, applying **anomaly detection** and **clustering** techniques to estimate power efficiency, and built a rule-based recommendation system that identifies simultaneous appliance usage to help manage the community's energy consumption efficiently.
- Developed a **multi-agent simulation (Mesa)** where a manager agent recommends adjusting usage based on each household's consumption trend, and household agents, *ideal*, *enthusiastic*, or *non-enthusiastic*, decide whether to follow it. Ran 18 parallel scenarios across behavior types and community sizes to find the best-performing configuration.
- Integrated live electricity pricing and renewable energy data through the **posf.ro REST API** to calculate savings and rank the best alternative providers for each scenario. Visualized results with **Plotly** and applied a **Pareto-based analysis** to identify the strongest balance between savings and energy efficiency.

HelpScout Mailbox Analyzer | *Python, Flask, API, HTML/CSS, JavaScript, SQLite*

- Shipped a full-stack platform that automates monthly support-cost allocation for the support team, ingesting **thousands of conversations** from **3 HelpScout mailboxes** into a normalized **SQLite** schema (**20+ products**, **40+ tags** linked by foreign keys) and cutting manual processing time by **95%**.
- Built a **5-stage async pipeline** across **3 HelpScout mailboxes**, using **asyncio** and **ThreadPoolExecutor** (up to **30 concurrent requests**) with **Abstract Base Classes**, **Template Method**, **Singleton**, and **Mixin** patterns, skipping out-of-range pages to save API calls.
- Designed logic to detect who answered each conversation across **3 mailbox types**, using **regex parsing**, **tag-based lookups**, and **multiple inheritance** for the hybrid case, on top of an API layer with **OAuth2**, **rate limiting**, and **retry with backoff**.

CERTIFICATES

CCNA 2 Cisco Certificate @ Hackacademy Mar. 2025 – May 2025

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, C, TypeScript, JavaScript, HTML, CSS

Frameworks & Libraries: Scrapy, Scrapy-Redis, PySpark, Flask, TensorFlow, Keras, scikit-learn

Cloud & Big Data: Azure, Databricks, Spark, Power BI

Dev Tools: Git, GitHub

Databases: Microsoft SQL Server, MySQL, SQLite

Languages (Spoken): Romanian (Native), English (B2)

Other: Crawling, Scraping, Machine Learning, Agent-Based Modelling

EDUCATION

National University of Science and Technology Politehnica Bucharest Sept. 2022 – June 2026

Bachelor's Degree at Faculty of Automatic Control and Computer Science

- Undergraduate Teaching Assistant for Object-Oriented Programming